

2. Klausur PraMa

Gruppe A

Beispiel 1:

$$f: \mathbb{R}^3 \rightarrow \mathbb{R} \quad f(x, y, z) = 3z \cdot \sqrt{x^2 + y^2}$$

$$r = 5 \quad z \geq 0$$

Kugelkoordinaten:

$$\det J: r^2 \sin \Theta$$

$$x = r \cdot \cos \varphi \sin \Theta$$

$$y = r \cdot \sin \varphi \sin \Theta$$

$$z = r \cdot \cos \Theta$$

$$x^2 + y^2 = r^2 \cos^2 \varphi \sin^2 \theta + r^2 \sin^2 \varphi \sin^2 \theta$$

$$x^2 + y^2 = r^2 \cdot \sin^2 \theta \cdot (\underbrace{\cos^2 \varphi + \sin^2 \varphi}_1)$$

$$x^2 + y^2 = r^2 \cdot \sin^2 \theta$$

$$\sqrt{x^2 + y^2} = r \cdot \sin \theta$$

$$x \in [0, 5]$$

$$\varphi \in [0, 2\pi]$$

$$\theta \in [0, \frac{\pi}{2}]$$

$$f = \underbrace{3 \cdot r \cdot \cos \theta}_{3^{\frac{1}{2}}} \cdot \underbrace{r \cdot \sin \theta}_{\sqrt{x^2 + y^2}}$$

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^5 3r \cos \theta r \sin \theta \cdot \overbrace{r^2 \sin \theta}^{\det J} dr d\theta d\varphi$$

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^5 3r^4 \cos \theta \sin^2 \theta dr d\theta d\varphi =$$

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \left. \frac{3}{5} r^5 \cos \theta \sin^2 \theta \right|_0^5 d\theta d\varphi =$$

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \frac{3 \cdot 5^5}{6} \sin^2 \theta \, d\theta \, dy$$

$$\frac{3 \cdot 5^5}{5} \cdot \int_0^{2\pi} -\sin \theta$$

$$2) \quad \ddot{y} + 9y = 2\sin(3t)$$

$$a) \quad \ddot{y} + 9y = 0$$

$$\lambda^2 + 9 = 0$$

$$\lambda^2 = -9$$

$$\lambda = \sqrt{-9}$$

$$\lambda = \pm i\sqrt{9}$$

$$\lambda = \pm i \cdot 3$$

$$\{y_1, y_2\} = \{e^{i3t}, e^{-i3t}\}$$

$$y_h(t) = c_1 \cdot e^{i3t} + c_2 e^{-i3t}$$

Komplex

$$\text{Euler Formel: } e^{i\varphi} = \cos(\varphi) + i \cdot \sin(\varphi)$$

$$\{y_1, y_2\} = \{\sin(3t), \cos(3t)\}$$

$$y_h = c_1 \cdot \sin(3t) + c_2 \cdot \cos(3t)$$

Reel

Innere Resonanz:

man benötigt ein weiteres

Äußere Resonanz:
man benötigt ein weiteres +

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$$b) \ddot{y} + 9y = 2\sin(3t) + 0 \cdot \cos(3t)$$

Ansatz:

$$y_p = (A \cdot \sin(3t) + B \cdot \cos(3t)) t^2$$

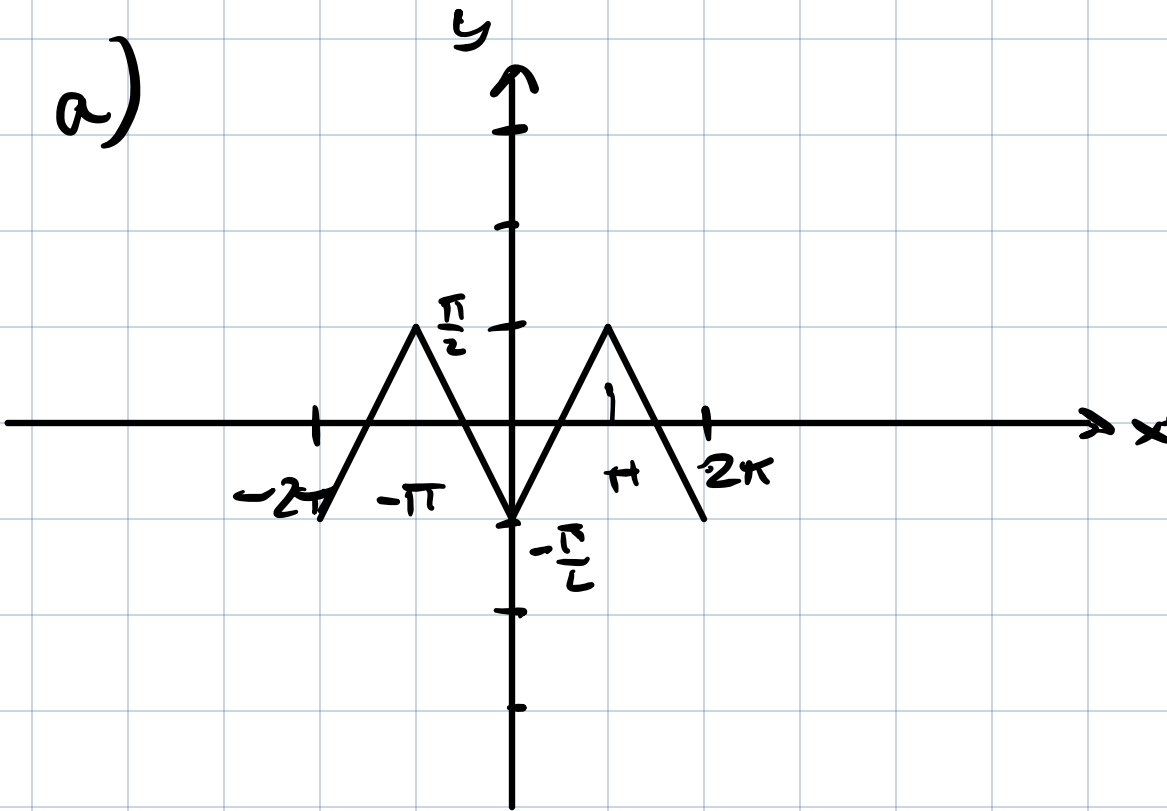
$$\dot{y}_p = (A \cdot \cos(3t) - B \cdot \sin(3t)) t^2 + 2t \cdot (A \sin(3t) + B \cos(3t))$$

$$\ddot{y}_p = (-A \sin(3t) - B \cos(3t)) t^2 + 2t (A \cos(3t) - B \sin(3t)) + 2t \cdot (A \cos(3t) - B \sin(3t)) + 2 \cdot (A \sin(3t) + B \cos(3t))$$

$$[-At^2 \sin(3t) - Bt^2 \cos(3t) + 4t(A \cos(3t) - B \sin(3t)) + 2A \sin(3t) + 2B \cos(3t)] = 9At^2 \sin(3t) + 9Bt^2 \cos(3t) = 2\sin(3t)$$

3) $f(x) = \begin{cases} x - \frac{\pi}{2} & 0 \leq x < \pi \\ -x + \frac{3\pi}{2} & \pi \leq x < 2\pi \end{cases}$ Daniel Kötler

a)



b) gerade Funktion:

$$b_k = 0$$

aus Skizze abgelesen:

$$a_0 = -\frac{\pi}{2}$$

$$a_k = \frac{1}{\pi} \cdot \int_0^{\pi} \left(x - \frac{\pi}{2}\right) \cos(kx) dx + \frac{1}{\pi} \cdot \int_{\pi}^{2\pi} \left(-x + \frac{3\pi}{2}\right) \cos(kx) dx$$

Fourier-Analyse:

$$f_n(x) = a_0 + \sum_{k=1}^n (a_k \cos(kx) + b_k \sin(kx))$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x) dx \quad a_k = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos(kx) dx \quad b_k = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(kx) dx$$

Trigonometrie:

$$\tan(x) = \frac{\sin(x)}{\cos(x)} \quad \cot(x) = \frac{\cos(x)}{\sin(x)}$$

$$\sin^2(x) + \cos^2(x) = 1 \quad \sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = 1 - 2\sin^2(x) \quad \cos(2x) = 2\cos^2(x) - 1$$

$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\sin x \cdot \sin y = \frac{1}{2} \cdot (\cos(x-y) - \cos(x+y))$$

$$\cos x \cdot \cos y = \frac{1}{2} \cdot (\cos(x-y) + \cos(x+y))$$

$$\sin x \cdot \cos y = \frac{1}{2} \cdot (\sin(x-y) + \sin(x+y))$$

$$\cosh^2(x) - \sinh^2(x) = 1$$

$$\sinh(x) = \frac{1}{2} \cdot (e^x - e^{-x})$$

$$\cosh(x) = \frac{1}{2} \cdot (e^x + e^{-x})$$

f	f'	f	F
$\sin x$	$\cos x$	0	C
$\cos x$	$-\sin x$	$\frac{1}{x}$	$\ln x $
$\tan x$	$\frac{1}{\cos^2 x}$	$\sin x$	$-\cos x$
$\cot x$	$-\frac{1}{\sin^2 x}$	$\cos x$	$\sin x$
$\ln x$	$\frac{1}{x}$	$\tan x$	$-\ln \cos x $
$\cosh x$	$\sinh x$	a^x	$\frac{a^x}{\ln(a)}$
$\sinh x$	$\cosh x$	$\ln(x)$	$x \cdot \ln x - x$
		$\log_a(x)$	$\frac{1}{\ln a} (x \ln x - x)$

Taylorpolynom:

$$\frac{f(a)}{0!} \cdot (x-a)^0 + \frac{f'(a)}{1!} \cdot (x-a)^1 + \frac{f''(a)}{2!} \cdot (x-a)^2 \dots$$

Komplexe Zahlen:

$$e^{i\varphi} = \cos(\varphi) + i \sin(\varphi)$$

Ableitungsregeln:

$$(f_1 \cdot f_2)' = f_1' \cdot f_2 + f_1 \cdot f_2'$$

$$\left(\frac{f_1}{f_2}\right)' = \frac{f_1' \cdot f_2 - f_1 \cdot f_2'}{f_2^2}$$

Integrationsregel:

$$\int a(x) b(x) dx = a(x) \cdot d - \int c \cdot d dx$$

$$a'(x) = c$$

$$\int b(x) dx = d$$

α	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	-
$\cot$	-	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0

$$\ln(1) = 0 \quad \log_a(x) = \frac{\ln(x)}{\ln(a)}$$

Leibniz-Regel:

$$g'(x) = \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x,y) dy + b'(x) \cdot f(x,b(x)) - a'(x) \cdot f(x,a(x))$$

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Formelzettel

Flächenintegrale:

$$A = \int_B 1 dF \quad (\text{bei Polar/Zylinder: } \det J = r \text{ dazu})$$

$$\underbrace{x^2 + y^2}_{r^2} \leq 1 \quad \dots \text{beschreibt Kreis}$$

Volumenintegrale:

$$V = \int_B 1 dV \quad (\text{bei Polar/Zyl./Kugel: } \det J = r/r/r^2 \sin \Theta \text{ dazu})$$

Koordinatentransformation:

Polar: $x = r \cdot \cos \varphi \quad y = r \cdot \sin \varphi$

Zylinder: $x = r \cdot \cos \varphi \quad y = r \cdot \sin \varphi \quad z = z$

Kugel: $x = r \cdot \cos \varphi \sin \Theta \quad y = r \cdot \sin \varphi \sin \Theta \quad z = r \cdot \cos \Theta$

Ellipsenfläche:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \dots x = a \cdot r \cdot \cos \varphi \quad y = b \cdot r \cdot \sin \varphi \quad \det J = abr$$

Diff. Gl. 1. Ord.

$$\dot{y} = f(x) \cdot y + g(x) \quad y_h = c \cdot e^{\int f(x) dx} \quad y_p = y_h \cdot \int \frac{g(x)}{y_h} dx$$

Diff Gl. 2. Ord.

$$y_h(t) = c_1 \cdot y_1(t) + c_2 \cdot y_2(t) \quad y_1(t) = e^{\lambda_1 t} \quad y_2(t) = e^{\lambda_2 t}$$

$$Y(t) = \begin{pmatrix} y_1 & y_2 \\ \dot{y}_1 & \dot{y}_2 \end{pmatrix} \Rightarrow \text{l.u. wenn } \det Y \neq 0 \quad (y_1 \cdot \dot{y}_2 - \dot{y}_1 \cdot y_2)$$

(Wronski-Det.)

$\{y_1, y_2\} \hat{=}$ Fundamentalsystem

$$\lambda = \alpha \pm i\beta$$

$$y(t) = e^{\alpha t} \cdot (c_1 e^{i\beta t} + c_2 e^{-i\beta t})$$
$$= y(t) = c_1 e^{\alpha t} \cdot \sin(\beta t) + c_2 e^{\alpha t} \cos(\beta t)$$

y_p : äußere Resonanz $\rightarrow t$ dazu (erst t^2)

keine äußere Resonanz $\rightarrow$ gar kein t

Inhomogenität f	Ansatz für y_p
$f(t) = b$	$\bar{y}_p(t) = a$
$f(t) = b \cdot e^{\mu t}$	$y_p(t) = a \cdot e^{\mu t}$
$f(t) = b_0 + b_1 t + \dots + b_n t^n$	$y_p(t) = a_0 + a_1 t + \dots + a_n t^n$
$f(t) = b_1 \sin(kt) + b_2 \cos(kt)$	$y_p(t) = a_1 \sin(kt) + a_2 \cos(kt)$
$f(t) = e^{\mu t} (b_1 \sin(kt) + b_2 \cos(kt))$	$y_p(t) = e^{\mu t} (a_1 \dots)$
$f(t) = e^{\mu t} p(t)$	$y_p(t) = e^{\mu t} q(t)$